

ANTI – DIABETIC DRUGS

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OVERVIEW

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- ▶ The pancreas produces hormones in the cluster of cells called islets of Langerhans which contain three major types of hormone secreting cells
- ▶ The alpha produces glucagon, the beta cells produce insulin and amylin while the delta cells secrete somatostatin
- ▶ Insulin promotes the uptake, usage and storage of glucose which leads to lowering the serum glucose levels
- ▶ Glucagon increases the hepatic glucose output and blood glucose and serum blood concentration
- ▶ Diabetes is a condition that is caused by inadequate insulin secretion or the reduced insulin activity (Insulin resistance) where insulin secreted is not sufficient to maintain normal blood

Insulin

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- ▶ Insulin peptide hormone is produced as pro insulin (Precursor) which is converted to insulin and C peptide
- ▶ Insulin consists of two peptide chains i.e. the A chain and the B chain which are linked by two disulphide bridges
- ▶ Although both insulin and C peptide are released in response to rising glucose concentrations, the physiologic role of C peptide remains unknown
- ▶ Insulin secretion has meal stimulated and basal components which is activated by the rise in blood glucose concentration that follows the digestion and absorption of carbohydrates
- ▶ Insulin is released at the rate of 1 IU per 10 kg of dietary carbohydrates and it peaks within 1 hour of eating which promotes uptake of carbohydrate and other nutrients uptake
- ▶ Both insulin and carbohydrate concentration return to pre-prandial concentration level within 2 hours
- ▶ Basal secretion which usually ranges from 0.5 to 1 IU of insulin per hour helps to retard hepatic glucose output during the absorptive state

Physiologic effects of insulin

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- ▶ Insulin is also referred to as “**storage hormone**” as it promotes formation of glycogen, triglycerides and protein inhibiting their breakdown
- ▶ The liver serves as the major source of blood glucose to supply the brain in the fasting state
- ▶ This supply line of glucose by the liver is done through two processes namely;
 - i. **Gluconeogenesis** – formation of glucose from amino acids
 - ii. **Glycogenolysis** – formation of glucose by through the breaking of glycogen
- ▶ Insulin stimulates enzymes involved in glycogen synthesis while inhibiting glycogenolytic and gluconeogenic enzymes thereby, reducing glucose output by the liver
- ▶ Generally insulin promotes the uptake of glucose by **skeletal muscle** and the **adipose tissue** by activating a glucose transporter in these tissues called GLUT 4.

Summary on the Synthesis and Release of Insulin

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In summary, Insulin release is moderated by the following;

- ▶ Glucose (most important) and ketone bodies stimulate release
- ▶ Glucagon and somatostatin inhibit release
- ▶ α -Adrenergic stimulation inhibits release (most important)
- ▶ β -Adrenergic stimulation promotes release
- ▶ Elevated intracellular Ca^{2+} promotes release

Glucagon

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- ▶ Glucagon is produced by alpha cells of the pancreas in response to decreased blood glucose concentrations
- ▶ It activates gluconeogenesis and glycogenolysis which increase hepatic glucose production
- ▶ Patients with diabetes continue to produce glucagon while there is a deficiency in insulin production and hence leading to an increase in serum glucose
- ▶ This imbalance contributes to the metabolic derangements of this disease

DIABETES MELLITUS

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Definition

- ▶ Diabetes is a heterogeneous group of syndromes characterized by the elevation of glucose concentration levels (basal and postprandial) due to a relative or absolute deficiency of insulin

- ▶ Symptoms of diabetes symptoms
 1. High level of glucose in urine and fasting blood
 2. Polyuria
 3. Polydypsia
 4. Fatigue
 5. Constant hunger

TYPES OF DIABETES

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TYPE 1 DIABETES MELLITUS: -

- ▶ This is a type of diabetes occurring in individuals with complete depletion of beta cells and hence causing the inability to produce insulin
- ▶ This destruction of functional pancreatic β -cells caused by T cell-mediated autoimmune response against the beta cells
- ▶ Lack of functional β -cells prevents mitigation of elevated glucose levels and associated insulin responses leading to excessive production of ketones and hence the occurrence of ketonemia and ketoacidosis
- ▶ Also referred to as “Childhood” diabetes which presents as sudden onset in younger age group i.e. before the age of 30, the median age being 15
- ▶ It is further known as insulin dependent diabetes and it is Usually controlled by insulin injections

TYPE 2 DIABETES

- ▶ This is the type of diabetes where there are residual beta cells which are able to produce insulin but not adequate to cater for the body's demand
- ▶ In other instances, insulin levels are normal or elevated but there is either a decrease in number of insulin receptors or the cells cannot take it up (insulin resistance)
- ▶ Insulin resistance is usually caused by the presence of insulin antibodies or defects in insulin receptors and signal transduction in the target organs
- ▶ Also referred to "Adult" diabetes as it usually occurs in individuals above the age 30 and it is often associated with a significant degree of insulin resistance and obesity
- ▶ Type 2 diabetes patients are less likely to develop ketonemia or ketoacidosis compared to the type 1 diabetes
- ▶ Most patients of type 2 diabetes either have normal or elevated levels of insulin and so do not need exogenous insulin but can be controlled by dietary changes , regular exercise oral hypoglycemic drugs

TREATMENT

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Type I DM

- ▶ Depend on exogenous insulin to prevent hyperglycemia and subsequent avoidance of ketoacidosis
- ▶ The goal of therapy is to mimic both the basal and reactive secretion of insulin in response to glucose levels avoiding both hyper- and hypo-glycemic episodes

Type II DM

- ▶ The goal of treatment is to maintain glucose concentrations within normal limits to prevent long term complications
- ▶ Weight reduction, exercise and dietary modification decrease insulin resistance essential steps in the treatment regimen
- ▶ Addition of hypoglycemic agents is often required
- ▶ Less often, insulin therapy is also required

Pharmacokinetics of insulin

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- ▶ Elimination half life ($t_{1/2}$) of IV insulin is 5 to 10 minutes
- ▶ Metabolized by kidneys and liver
- ▶ Insulin is tightly bound to tissue receptors and so even if it has rapid clearance, its pharmacologic effects remain till 30 - 60 minutes
- ▶ Insulin is secreted into portal venous system at the rate of 1 unit/hour
- ▶ Daily secretion of insulin is about 40 units
- ▶ Insulin response to glucose is greater for oral ingestion than for iv infusion

INSULIN PREPARATIONS

VERY RAPID ACTING

- ▶ Lispro insulin, insulin asparte and Glulisine insulin (ultrashort acting)
- ▶ More rapid than regular insulin and shorter duration
- ▶ Less associated with hypoglycemia but recurrent hyperglycemia before next meal due to shorter duration of action
- ▶ Given just 15 minutes prior to meals - Provides a postprandial plasma insulin concentration like normal insulin

INTERMEDIATE ACTING

- ▶ Lente (NPH) - Absorption delayed due to conjugation with protamine (0.005mg/unit)

LONG ACTING

- ▶ Ultralente, Glargine
- ▶ Given as single bedtime injection
- ▶ Presents lesser nocturnal hypoglycemia

ADVERSE EFFECTS OF INSULIN

- ▶ Hypoglycemia (Headache, anxiety, tachycardia, confusion, vertigo, diaphoresis, shaky, increased appetite, blurred vision and weakness or fatigue)
- ▶ Allergic reactions (Hypersensitivity)
- ▶ Lipodystrophy (Local atrophy or hypertrophy of subcutaneous fatty at site of injection)
- ▶ Insulin resistance
- ▶ Hypokalemia
- ▶ Weight gain

SYNTHETIC AMYLIN ANALOGUES

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▶ **Example** : Pramlintide

▶ Amylin is a hormone that is co-secreted with insulin from β cells following food intake

Mechanism of action

▶ It delays gastric emptying

▶ Decreases postprandial glucagon secretion and improves satiety (a feeling of fullness)

Indication

▶ Pramlintide is indicated as an adjunct to mealtime insulin therapy in patients with type 1 and type 2 diabetes

▶ It is administered by subcutaneous injection immediately prior to meals

Side effects of pramlintide

- ▶ Hypoglycemia – thus on initiating treatment, the dose of mealtime insulin should be decreased by 50% to avoid a risk of severe hypoglycemia
- ▶ Nausea
- ▶ Anorexia
- ▶ Vomiting

Caution

- ▶ Pramlintide may not be mixed in the same syringe with insulin
- ▶ It should be avoided in patients with diabetic gastroparesis (delayed stomach emptying), cresol hypersensitivity or hypoglycemic unawareness

INCRETIN MIMETICS

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- ▶ **Examples** : Exenatide, liraglutide
- ▶ Oral glucose results in a higher secretion of insulin than occurs when an equal load of glucose is given IV
- ▶ This effect is referred to as the “**incretin effect**” and is markedly reduced in type 2 diabetes
- ▶ The incretin effect occurs because the gut releases incretin hormones, notably glucagonlike peptide-1 (GLP-1) and glucose-dependent insulintropic polypeptide, in response to a meal
- ▶ Incretin hormones are responsible for 60% to 70% of postprandial insulin secretion
- ▶ Incretin mimetics are administered as injectable and used for the treatment of type 2 diabetes

Mechanism of action of incretin mimetics

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- ▶ The incretin mimetics are analogues of GLP-1 that exert their activity by acting as GLP-1 receptor agonists
- ▶ These agents improve glucose dependent insulin secretion
- ▶ Slow gastric emptying time
- ▶ Reduce food intake by enhancing satiety
- ▶ Decrease postprandial glucagon secretion
- ▶ Promote β -cell proliferation
- ▶ Consequently, weight gain and postprandial hyperglycemia are reduced, and HbA1c levels decline

Pharmacokinetics of incretin mimetics

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- ▶ Being polypeptides, exenatide and liraglutide must be administered subcutaneously
- ▶ Liraglutide is highly protein bound and has a long half-life, allowing for once-daily dosing without regard to meals
- ▶ Exenatide is eliminated mainly via glomerular filtration and has a much shorter half-life
- ▶ Because of the short duration of action, exenatide should be injected twice daily within 60 minutes prior to morning and evening meals
- ▶ A once-weekly extended-release preparation is also available
- ▶ **NB:** Exenatide should be avoided in patients with severe renal impairment

Side effects of incretin mimetics

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- ▶ The main adverse effects of the incretin mimetics consist of
 - ▶ Nausea
 - ▶ Vomiting
 - ▶ Diarrhea
 - ▶ Constipation
- ▶ Exenatide and liraglutide have been associated with pancreatitis and so patients should be advised to discontinue these agents and contact their health care provider immediately if they experience severe abdominal pain
- ▶ While Liraglutide causes thyroid C-cell tumors in rodents, it is unknown if it causes these tumors or thyroid carcinoma in humans

ORAL HYPOGLYCEMICS

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- ▶ These agents are useful in the treatment of Type II DM where there is no adequate response to non-medical interventions (diet, exercise and weight loss)
- ▶ Newly diagnosed Type II DM (less than 5 years) often respond well to oral agents
- ▶ Patients with long standing disease (often diagnosed late) require a combination of agents with or without insulin
- ▶ The progressive decline in β -cell function often necessitates the addition of insulin at some time in Type II DM
- ▶ Oral agents are never indicated for Type I DM

SULFONYLUREAS

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- ▶ Drugs in this class are categorized according to generations to which they belong
- ▶ They are further classified according to their onset and duration of action

1. Classification according to generations

First generation

- ▶ Tolbutamide, Chlorpropamide, Acetohexamide, Tolazemide

Second generation

- ▶ Glibenclamide, glipizide, gliclazide, Glimepyride

2. Classification according to onset of action

- ☐ Short acting: Tolbutamide
- ☐ Intermediate acting: Glipizide ,Glyburide
- ☐ Long acting: Chlorpropamide, Glimepride
- ▶ Sulphonyrueleas promote the release of insulin from β -cells and hence they are also referred as **insulin secretagogues**

Mechanism of action of Sulphonyrleas

- ▶ These agents require functioning β -cells for them to enhance secretion of insulin
- ▶ They stimulate release of insulin by blocking ATP-sensitive K^+ channels resulting in depolarization with Ca^{2+} influx which promotes insulin secretion
- ▶ They also reduce glucagon secretion and increase the binding of insulin to target tissues
- ▶ They may also increase the number of insulin receptors

Adverse Effects of sulphonylureas:

- ▶ Weight gain, Hyperinsulinemia and Hypoglycemia
- ▶ Hepatic or renal insufficiency causes accumulation of these agents promoting the risk of hypoglycemia
- ▶ Tolbutamide is associated with a 2.5 times rise in cardiovascular mortality
- ▶ Hypersensitivity reaction to those allergic to Sulphur containing drugs

MEGLITINIDES

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- ▶ Examples are
 - Repaglinide
 - Nateglinide
- ▶ They also act as secretagogues
- ▶ Prompt peak effect (1 hour) and shorter duration of action (about 4 hours)
- ▶ Taken 15-30 minutes before meals
- ▶ Decreases risk of prolonged hypoglycemia due to short duration of action
- ▶ When used in combination with other oral agents they produce better control than any monotherapy

BIGUANIDES

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▶ Examples

- Metformin
- ▶ Classified as an insulin sensitizer- increases glucose uptake and utilization by target tissues

Mechanism of action

- ▶ Reduces plasma glucose levels by inhibiting hepatic gluconeogenesis and glycogenolysis
- ▶ It also reduces hyperlipidemia (↓LDL and VLDL cholesterol and ↑ HDL)
Lipid lowering requires 4-6 weeks of treatment
- ▶ Metformin increases the glucose uptake by peripheral tissues
- ▶ Metformin also decreases appetite
- ▶ It is the only oral hypoglycemic shown to reduce cardiovascular mortality

Adverse effects

- ▶ Hypoglycemia occurs only when combined with other agents
- ▶ Rarely severe lactic acidosis is associated with metformin use particularly in diabetics with CHF
- ▶ Drug interactions with cimetidine, furosemide, nifedipine and others have been identified.
- ▶ Reduces the intrinsic factor responsible for vitamin 12 absorption and hence can predispose an individual to megaloblastic anemia

THIAZOLIDINEDIONES (GLITAZONES)

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- ▶ Examples
 - Pioglitazone
 - Rosiglitazone
- ▶ These drugs are referred to as insulin sensitizers and as shown above, they are also called glitazones
- ▶ They do not promote insulin secretion from β -cells but insulin is necessary for them to be effective

Mechanism of action of thiazolidinediones

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- ▶ Act principally at the adipose tissue and skeletal muscles to decrease insulin resistance
- ▶ This is achieved through their selective agonist effect at the peroxisome proliferator activated receptor gamma within the target tissues for insulin action like the adipose and the liver
- ▶ This then leads to the increase in the transcription of insulin responsive genes involved in control of glucose and lipid production
- ▶ This all eventually leads to tissue sensitivity to insulin while reducing hepatic production of glucose
- ▶ Clinical effect takes 4 -12 weeks to be evident

Side effects

- ▶ Weight gain
- ▶ Can induce hepatic dysfunction

ALPHA-GLUCOSIDASE INHIBITORS

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Examples

- Acarbose
- Miglitol

Mechanism of action

- ▶ These agents are oligosaccharide derivatives taken at the beginning of a meal
- ▶ Delay carbohydrate digestion by competitively inhibiting α -glucosidase, a membrane bound enzyme of the intestinal brush border

Adverse Effects of alpha glucosidase inhibitor

- ▶ Flatulence
- ▶ Diarrhea
- ▶ Cramping

Interactions

- ▶ Metformin bioavailability is severely decreased when used concomitantly
- ▶ These agents should not be used in diabetics with intestinal pathology

DIPEPTIDYL PEPTIDASE-4 (DPP 4) INHIBITORS

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- ▶ Examples: Alogliptin, Linagliptin, Saxagliptin, Sitagliptin

Mechanism of action

- ▶ These drugs inhibit the enzyme DPP-4, which is responsible for the inactivation of incretin hormones such as GLP-1
- ▶ Prolonging the activity of incretin hormones increases insulin release in response to meals and reduces inappropriate secretion of glucagon
- ▶ DPP-4 inhibitors may be used as monotherapy or in combination with sulfonylureas, metformin, TZDs, or insulin
- ▶ Unlike incretin mimetics, these drugs do not cause satiety, or fullness, and are weight neutral

Pharmacokinetics of DPP-4 inhibitors

- ▶ The DPP-4 inhibitors are well absorbed after oral administration
- ▶ Food does not affect the extent of absorption
- ▶ Alogliptin and sitagliptin are mostly excreted unchanged in the urine
- ▶ Saxagliptin is metabolized via CYP450 3A4/5 to an active metabolite
- ▶ The primary route of elimination for saxagliptin and the metabolite is renal
- ▶ Linagliptin is primarily eliminated via the enterohepatic system
- ▶ All DPP-4 inhibitors except linagliptin require dosage adjustments in renal dysfunction

Side effects of DPP-4 inhibitors

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- ▶ In general, DPP-4 inhibitors are well tolerated, with the most common adverse effects being nasopharyngitis and headache
- ▶ Although infrequent, pancreatitis has occurred with use of all DPP-4 inhibitors
- ▶ Strong inhibitors of CYP450 3A4/5, such as ritonavir, atazanavir, itraconazole, and clarithromycin, may increase levels of saxagliptin
- ▶ Therefore, reduced doses of saxagliptin should be used

SODIUM GLUCOSE CO-TRANSPORTER 2 INHIBITORS

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Examples

- ▶ Canagliflozin, Dapagliflozin, Empagliflozin, Jardiance

Mechanism of action

- ▶ The sodium-glucose co-transporter 2 (SGLT2) is responsible for reabsorbing filtered glucose in the tubular lumen of the kidney
- ▶ By inhibiting SGLT2, these agents decrease reabsorption of glucose, increase urinary glucose excretion, and lower blood glucose
- ▶ Inhibition of SGLT2 also decreases reabsorption of sodium and causes osmotic diuresis
- ▶ Therefore, SGLT2 inhibitors may reduce systolic blood pressure though not indicated for the treatment of hypertension

Pharmacokinetics of SGLT inhibitors

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- ▶ These agents are given once daily in the morning
- ▶ Canagliflozin should be taken before the first meal of the day
- ▶ Both drugs are mainly metabolized by glucuronidation to inactive metabolites
- ▶ While the primary route of excretion for canagliflozin is via the feces, about one-third of a dose is renally eliminated
- ▶ These agents should be avoided in patients with renal dysfunction

Adverse effects

- ▶ The most common adverse effects with SGLT2 inhibitors are female genital mycotic infections (for example, vulvovaginal candidiasis), urinary tract infections, and urinary frequency
- ▶ Hypotension has also occurred, particularly in the elderly or patients on diuretics
- ▶ Thus, volume status should be evaluated prior to starting these agents

END